

Density-Activated Dark Photon as the Gauge-Sector Realisation of the NVG Magnetar Mechanism

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Repository: <https://github.com/infosave2007/vmf>

Abstract

The V2 magnetar preprint of this series identifies a single dimensionless coupling $\lambda \sim 0.9$ at $\rho \sim 2n_0$ as the residual non-perturbative parameter that controls the structural amplification factor $\Gamma_{\text{struct}} \sim 2\text{--}3$. The companion derivation in `article/NVG_EM_RESPONSE_DERIVATION.md` showed that aggregating every standard-QCD operator that couples the chiral condensate to the photon kinetic term (quark QED loop, charged-pion loop, vector-meson dominance, QCD magnetic susceptibility) gives $\lambda_{\text{std}} \sim 4 \times 10^{-3}$, a shortfall of two orders of magnitude. The mechanism therefore cannot be a pure-QCD effect.

This note proposes and explores the most economical beyond-minimal-QCD realisation consistent with the NVG framework: a hidden $U(1)$ gauge field A'_μ kinetically mixed with the visible photon, with mixing parameter $\varepsilon(\rho) = \varepsilon_0 (1 - \sigma^2(\rho)/\sigma_0^2)$ activated by the chiral order parameter. We show that

- the V2 benchmark $\lambda = 0.9$ corresponds to $\varepsilon_0 \approx 1.0$ with $m_{A'} \sim \Lambda_\chi \sim 1.16$ GeV;
- within the simplified benchmark scan implemented here, this ansatz is not immediately excluded by representative astrophysical, cosmological, and laboratory bounds on kinetic mixing (Sun, HB stars, red giants, BBN, beam dumps, BaBar, SN1987A) by virtue of two independent suppressions: density activation $S(\rho) \approx 0$ outside dense matter, and kinematic blockade $m_{A'} > T_{\text{env}}$ in all stellar/cosmological environments;
- the resulting model produces five qualitatively distinct predictions that separate density-activated mixing from standard constant- ε dark photons.

The numerical implementation, full bound table, and reproduction script are in `verification/nvg_dark_photon_k`

1. Setting and Motivation

The standard-QCD upper bound on the chiral-condensate-photon coupling (companion note `article/NVG_EM_RESPONSE_DERIVATION.md`, Sec. 5a) leaves the V2 magnetar mechanism in a sharp dichotomy. Either the NVG channel contributes only a sub-percent correction to the magnetar field ($\Gamma_{\text{struct}} \approx 1.002$), in which case it cannot resolve the magnetar paradox on its own; or the required coupling $\lambda \sim 0.9$ must be carried by a new operator in the gauge sector. The NVG framework naturally admits the latter, since the NVG vacuum sector by construction modifies the structure of gauge-field renormalisation in dense matter.

The minimal, falsifiable realisation explored here is a hidden $U(1)_d$ gauge field A'_μ that kinetically mixes with the visible photon. The crucial NVG ingredient is that the mixing parameter is *not* a constant of nature: it is set by the local NVG order parameter, equivalently the chiral condensate

$\sigma(\rho) = \langle \bar{q}q \rangle$. This single density dependence is what reconciles a near-order-unity mixing in dense matter with the stringent stellar/cosmological bounds on constant- ε dark photons.

2. The Model

The Lagrangian, restricted to the gauge sector, is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{4}F'_{\mu\nu}F'^{\mu\nu} - \frac{\varepsilon(\rho)}{2}F_{\mu\nu}F'^{\mu\nu} - \frac{1}{2}m_{A'}^2 A'_\mu A'^\mu,$$

with the density-dependent mixing

$$\boxed{\varepsilon(\rho) = \varepsilon_0 S(\rho), \quad S(\rho) \equiv 1 - \frac{\sigma^2(\rho)}{\sigma^2(0)}}.$$

$S(\rho)$ vanishes in the chirally broken vacuum ($\sigma \rightarrow \sigma_0$) and approaches unity at full chiral restoration ($\sigma \rightarrow 0$). In the NJL solution used throughout this series, $S(\rho = 2n_0) \approx 0.903$.

The two parameters of the model are the bare mixing ε_0 and the hidden-photon mass $m_{A'}$. We argue that NVG naturally selects $m_{A'} \sim \Lambda_\chi = 4\pi f_\pi \approx 1.16$ GeV (the chiral symmetry breaking scale, which is the only natural scale in the problem once we exclude the perturbative QED scale and the NS plasma scale).

The connection to the V2 EFT coupling is, in the regime $m_{A'} \gg \omega_p(\text{NS})$,

$$\delta Z_\chi(\rho) \approx \varepsilon^2(\rho) = \varepsilon_0^2 S^2(\rho).$$

Matching to the V2 ansatz $Z_\chi = 1 - \lambda(1 - r^2) = 1 - \lambda S(\rho)$ at the benchmark density yields

$$\varepsilon_0^2 = \frac{\lambda}{S(\rho = 2n_0)} \implies \varepsilon_0 \approx 1.0 \quad \text{for } \lambda \approx 0.9.$$

3. Illustrative Compatibility with Existing Bounds

The script enumerates seven representative environments spanning 30 orders of magnitude in baryon density and 16 in temperature. For each environment we compute the density-suppression factor $S(\rho)$, the effective in-medium mixing $\varepsilon_{\text{eff}}(\rho) = \varepsilon_0 S(\rho)$, and the kinematic threshold for producing on-shell A' pairs given the local temperature. This is an illustrative consistency scan, not a full global recast of the dark-photon exclusion literature.

Reading the script's output (frozen in `verification/nvg_dark_photon_kinetic_mixing_output.txt`):

Environment	ρ/n_0	$S(\rho)$	ε_{eff}	Bound on plain ε	Status
Vacuum lab (BaBar, NA64)	0	0	0	10^{-3}	ALLOWED ($S = 0$)
Sun	1.3×10^{-13}	0	n/a	10^{-14}	KINEMATIC-OFF
HB He core	4×10^{-11}	0	n/a	10^{-13}	KINEMATIC-OFF

Environment	ρ/n_0	$S(\rho)$	ε_{eff}	Bound on plain ε	Status
RG core	6×10^{-10}	0	n/a	3×10^{-13}	KINEMATIC-OFF
BBN plasma	10^{-30}	0	n/a	10^{-9}	KINEMATIC-OFF
SN1987A proto-NS	1.0	0.39	n/a	10^{-9}	KINEMATIC-OFF
Magnetar interior	2.0	0.90	0.90	(no bound)	TARGET

Within this benchmark setup, two distinct mechanisms suppress the listed bounds:

- (i) **Density activation.** For all stellar and cosmological environments the chiral condensate is essentially at its vacuum value, so $S \rightarrow 0$ and $\varepsilon_{\text{eff}} \rightarrow 0$ independently of ε_0 .
- (ii) **Kinematic blockade.** For $m_{A'} \approx 1.16$ GeV no environment with $T \lesssim 100$ MeV can produce on-shell A' pairs. This includes SN1987A ($T \sim 30$ MeV), where standard dark-photon bounds become much weaker for $m_{A'} > 200$ MeV and vanish entirely above the proton mass (Chang, Essig, McDermott 2017).

In this simplified table the benchmark point yields zero exclusions. The only “borderline” entry is the magnetar interior itself, which is the target environment where no observational bound on ε exists because the mechanism has never been searched for there. This should be read only as an existence proof that a dense-matter-activated parameter region is not obviously ruled out; a publication-grade exclusion study would still require a dedicated recast of each bound in the density-dependent model.

4. Distinguishing Predictions

Density-activated kinetic mixing is not merely a relabelling of standard dark photons. In the present benchmark it leads to the following qualitative phenomenology:

1. **Strong suppression in vacuum-based laboratory searches.** Beam dumps (E137, NA64), B -factory dark-photon searches (BaBar, Belle II, LHCb), and haloscope-style direct dark-photon detection all probe ε at $\rho \approx 0$. Since $\varepsilon(0) = 0$ in the model ansatz, the leading on-shell kinetic-mixing signal is strongly suppressed in such environments.
2. **Strong suppression in sub-nuclear stellar cooling channels.** Solar neutrino counts, HB-star lifetimes, RG-tip luminosities, and white-dwarf cooling ages should show no deviation from Standard Model predictions, again at leading order in this ansatz, since $\rho \ll n_0$ in all these environments.
3. **A possible magnetar B-field amplification correlated with central density.** Γ_{struct} scales monotonically with $S(\rho_c)$, so heavier (denser) magnetars should display systematically larger inferred surface fields, after correction for age. The population catalogue analysed in V2 (32/32 sources) is consistent with this trend but does not yet test it differentially.
4. **A possible dilepton signature in heavy-ion collisions at $\sqrt{s_{NN}} \sim \text{few GeV}$.** FAIR/NICA-energy collisions reach baryon chemical potentials $\mu_B \sim 1$ GeV during the dense fireball phase, where $S(\mu_B)$ can become $\mathcal{O}(1)$ for a brief window. A' production via $\bar{q}q \rightarrow A' \rightarrow \ell^+\ell^-$ could show as an excess dilepton signal in the $M_{\ell\ell} \sim m_{A'} \sim 1.16$ GeV bin,

scaling with the dense-phase lifetime, if the dense phase is long enough and the in-medium width is not too large.

5. **Suppressed EM precursor before NS–NS merger; possible sharp amplification within ~ 1 ms of contact.** The pre-merger inspiral densities are sub-nuclear (tidal disruption regime), so $\varepsilon_{\text{eff}} \approx 0$ and any kinetic-mixing-driven EM precursor is correspondingly suppressed. After contact the central density rapidly exceeds n_0 and the mechanism activates, contributing to the brightness and time profile of the EM counterpart in the first millisecond. This is qualitatively different from models with constant ε , which predict precursor signals during the inspiral.

Predictions (4) and (5) are, in principle, testable on a multi-year timescale with existing or near-future facilities (HADES, STAR-BES, CBM at FAIR; LIGO-Virgo-KAGRA O5), but extracting them would require a dedicated detector-level analysis that is outside the scope of the present note.

5. What This Does Not Solve

The model deliberately introduces *new physics*: a hidden gauge field with $\mathcal{O}(1)$ bare kinetic mixing controlled by the NVG vacuum sector. This is honest BSM, not a derivation from the Standard Model.

The remaining gaps:

- The NVG-side coupling $\varepsilon_0(\mathcal{W})$ has been postulated, not derived from the NVG action. A microscopic derivation of why the NVG order parameter $\mathcal{W}$ sources hidden-sector kinetic mixing is necessary for the framework to be fully self-contained.
- The hidden photon mass $m_{A'} \sim \Lambda_\chi$ is motivated only by scale matching. A first-principles calculation of $m_{A'}$ in terms of NVG sector parameters would close the second free parameter.
- The static-limit approximation $\delta Z_\chi \approx \varepsilon^2$ ignores plasma effects in the NS interior. A full in-medium photon dispersion calculation may shift ε_0 by $\mathcal{O}(1)$.

None of these gaps changes the central limited conclusion of this note: an $\mathcal{O}(1)$ bare kinetic mixing with density activation remains a plausible exploratory realisation of the V2 magnetar mechanism that is not obviously excluded by the representative constraints considered here.

6. Reproducing the Numbers

The full numerical analysis is implemented in `verification/nvg_dark_photon_kinetic_mixing.py` (~280 lines, no external dependencies beyond the in-repo NJL solver `verification/nvg_em_response_derivation`). Running

```
python3.12 verification/nvg_dark_photon_kinetic_mixing.py
```

reproduces the bound table of Sec. 3 and the benchmark mapping $\varepsilon_0 \approx 1.0$, $m_{A'} \approx 1.16$ GeV.

The frozen output is `verification/nvg_dark_photon_kinetic_mixing_output.txt`.

References

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8. NVG companion EM-response derivation: `article/NVG_EM_RESPONSE_DERIVATION.md`.